EE3-23 MACHINE LEARNING EXAM PAPER 2018

Formulas that may be needed:

$$\mathbb{P}\left(|\widehat{R}_n(h) - R(h)| > \varepsilon\right) \leq 4n^{d_{VC}} e^{-\varepsilon^2 n/8}$$

$$\theta(s) = \frac{e^s}{1+e^s}, \qquad \theta(s) = \max\{0, s\}$$

$$\theta(s) = \tanh(s) = \frac{e^s - e^{-s}}{e^s + e^{-s}}, \qquad \frac{\partial \tanh(s)}{\partial s} = 1 - \tanh^2(s)$$

1. Practical ML scenario.

   a) HMRC is considering using ML to identify suspicious tax return cases. Overall, it estimates about $£4.4 \cdot 10^9$ in taxes is lost due to tax evasion each year. The average cost of investigating a taxpayer is approximately $£10^4$. There are approximately 10 million taxpayers submitting their own tax returns, which are in structured form consisting of 100 fields, that can be converted into real value numbers. There are approx 4400 tax evasion cases every year and their records from the past 10 years are available. HMRC will find ML useful if it can guarantee that the test error will not differ from training by more than 20% with 99% certainty.

      i) Identify relevant ML components and formulate it as an ML problem. [30 marks]

      ii) What is required to guarantee that the predictor meets HMRC criteria? [10 marks]

   b) You decided to use a non-linear transformation of the input data with polynomials. You observe that the training error is zero but the test error is much larger.

      i) Give an example of a target function $f(x) =?$, polynomial feature transformation $\mathbf{z} = \Phi(x) = \{\ldots\}?$ and error function $\widehat{R} =?$ for the given scenario. In what specific training setting this may occur and how to improve the test error? [20 marks]

ii) Assume you fix the polynomial degree to $k = 3$. What is the approximate number of data points that are needed if you want to guarantee that the test error of the predictor is within 0.1 from the training error with probability 0.99? Show your calculations. [10 marks]

2. Consider neural network architecture given in Figure 2.

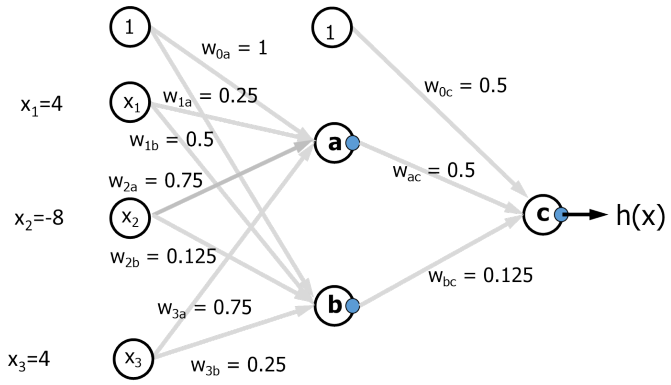


Figure 2.1 Neural network architecture

a) Forward propagation

i) Propose a non-linearity activation function, draw this function and its gradient in a figure. [10 marks]

ii) Discuss what happens if we use $\mathbf{x}$ as input, $\theta(s) = s$ as the activation function and W_l as a matrix of weights in layer l. Can $\theta(s) = s$ be achieved with $\tanh(s)$? [15 marks]

iii) Given input vector $\mathbf{x} = (4, -8, 4)$ use the proposed non-linearity from question i) to calculate outputs of all neurons and $h(x)$. [15 marks]

b) Backpropagation

i) Write down the main steps of the backpropagation algorithm. [25 marks]

ii) What is the role of the learning rate in gradient descent and what are the risks of setting the learning rate too large or too small? [10 marks]

iii) Apply the backpropagation algorithm to calculate the update of weight $w_{3,b}^{(2)}$. Use L_2 loss without regularization, ReLU activation, learning rate of 0.1 and training example of $\mathbf{x} = (4, -8, 4)$ with its label $y = 2$. [25 marks]

3. Support Vector Machines

a) You are given the following training data examples that belong to two classes:

Positive examples: $\begin{bmatrix}-3\\2\end{bmatrix}, \begin{bmatrix}-1\\1\end{bmatrix}, \begin{bmatrix}-1\\-1\end{bmatrix}, \begin{bmatrix}1\\-3\end{bmatrix}, \begin{bmatrix}2\\-2\end{bmatrix}$

Negative examples: $\begin{bmatrix}2\\4\end{bmatrix}, \begin{bmatrix}3\\1\end{bmatrix}, \begin{bmatrix}5\\1\end{bmatrix}, \begin{bmatrix}1\\3\end{bmatrix}$

i) Plot these points, and draw the maximum margin SVM classifier. Which of the training examples constitute the support vectors? [16 marks]

ii) The hyperplane $H(w,b)$ for this SVM classifier is defined as $w^T x + b = 0$, where $\|w\|^2 = 1$. Characterize the w and b parameters for the above training data. [12 marks]

iii) Suppose you are given the following additional training examples:

Positive example: $\begin{bmatrix}0\\-2\end{bmatrix}$ Negative examples: $\begin{bmatrix}3\\3\end{bmatrix}, \begin{bmatrix}5\\0\end{bmatrix}$

What are the w and b parameters of the SVM classifier now? [8 marks]

b) Consider the soft margin SVM where we penalize the sum of the l_2 norms of the margin violations (instead of l_1 norm as done in the class). We now need to solve the following optimization problem:

$$\text{minimize} \qquad \tfrac{1}{2}\|w\|^2 + \tfrac{C}{2}\textstyle\sum_{i=1}^{n} \xi_i^2$$

$$\text{subject to} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i \quad \text{and} \quad \xi_i \geq 0 \text{ for all } i$$

i) Show that we can remove the $\xi_i \geq 0$ constraints without loss of optimality; that is, the solution is the same without these constraint. [12 marks]

ii) Write down the Lagrangian, $\mathscr{L}(w,b,\xi,\alpha,\beta)$, for this optimization problem, where $\xi = [\xi_1,\ldots,\xi_n]^T$, $\alpha = [\alpha_1,\ldots,\alpha_n]^T$, $\beta = [\beta_1,\ldots,\beta_n]^T$, and α_i denotes the Lagrange multiplier for the inequality constraint $y_i(w^\top x_i + b) \geq 1 - \xi_i$, while β_i is the Lagrangian multiplier for the constraint $\xi_i \geq 0$. [12 marks]

iii) Minimize the Lagrangian with respect to w, b, and ξ by taking the

gradients, $\nabla_w \mathscr{L}$, $\nabla_b \mathscr{L}$, and $\nabla_\xi \mathscr{L}$, and then equating them to 0. [20 marks]

iv) What is the dual of this alternative soft-margin SVM optimization problem? [20 marks]

4. Consider the Markov decision process (MDP) with three states, A, B and C. The state transitions and rewards for all possible actions at each state are illustrated in Figure 4.1. In the figure, lower case letters a1, a2, b, c1, c2 denote actions, signed integers denote rewards, and the fractions denote transition probabilities.

For example, at state C, if we take action c1, we receive an immediate reward of $+8$ and transition to state A; whereas with action c2, we receive an immediate reward of $+2$, and transition to state B with probability 0.5 and remain in state C with probability 0.5.

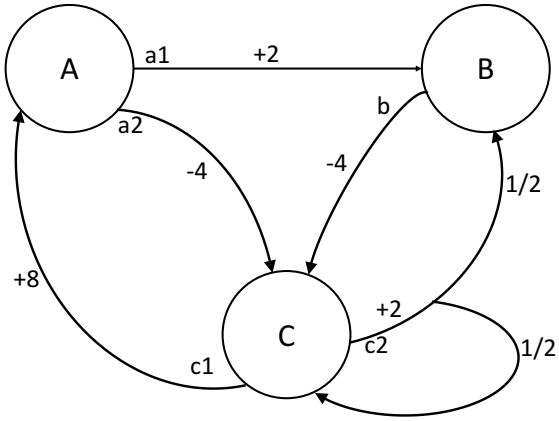


Figure 4.1 MDP with three states, and the corresponding state transitions and rewards.

a) Define the state-value function $v_\pi(s)$ for a discounted MDP. [12 marks]

b) Write down the Bellman equation for the state-value function. [12 marks]

c) Consider the uniform random policy $\pi_1(s,a)$ which takes all actions at any state s with equal probability. Start with initial value functions $v_1(A) = v_1(B) = v_1(C) = 4$, apply one synchronous iteration of policy evaluation with discount factor $\lambda = 0.5$ to compute a new value function $v_2(s)$. [16 marks]

d) Apply one iteration of the greedy policy improvement to compute a new deterministic policy $\pi_2(s)$. [12 marks]

e) Define the optimal state-value function $v^*(s)$ for an MDP. [8 marks]

f) Write down the Bellman optimality equation for state-value functions. [12 marks]

g) Starting with an initial value function of $v_1(A) = v_1(B) = v_1(C) = 4$, and $\lambda = 0.5$, apply one synchronous iteration of the value iteration algorithm to compute a new value function $v_2(s)$. [16 marks]

h) Is the value function $v_2(s)$ obtained above optimal? Explain your answer. [12 marks]